Screenshot :

[1]
23s

```
!pip install pyspark

from pyspark.sql import SparkSession

spark = SparkSession.builder \
.appName("Distributed Data Processing Activity") \
.getOrCreate()
```

```
Requirement already satisfied: pyspark in /usr/local/lib/python3.12/dist-packages (4.0.2)
Requirement already satisfied: py4j<0.10.9.10,>=0.10.9.7 in /usr/local/lib/python3.12/dist-packages (from pyspark) (0.10.9.9)
```

[5]
17s

```
from pyspark.sql.functions import col
from google.colab import drive

drive.mount('/content/drive')

file_path = '/content/drive/MyDrive/PDC_CFILES/student_mental_health_burnout_1M.csv'


df = spark.read.csv(file_path, header=True, inferSchema=True)

df.show(5)
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
+---+------+-------------+------------------+------------------+-------------------+------------------+------------------+------------------+------------------+------------------+------------------+------
|age|gender|academic_year|study_hours_per_day|     exam_pressure|academic_performance|      stress_level|     anxiety_score|  depression_score|      sleep_hours| physical_activity|    social_support|       screen_time|    
+---+------+-------------+------------------+------------------+-------------------+------------------+------------------+------------------+------------------+------------------+------------------+------
| 23|  Male|            2| 5.596070637712916| 6.487218156883566|   68.41111424333613|  4.116949660293958|2.2757133559191067|1.9867301997660676|6.880544690342919| 2.72886110154767| 6.47007979026500 3| 4.993800836186524|  4.9
| 20|  Male|            3|  5.597170756105616| 5.631480689038198|   67.68215874122792|0.34948942254822146|               0.0|               0.0|7.463339242884696|3.6908661999745864|               0.0|3.8629795303932193| 5.1
| 29|  Male|            2| 2.580491083921536 3| 6.015297104841322|   58.37236267885964|   3.476176844283 32|2.4252007031850 63|0.8519955424339019|8.946669952792416| 3.29672008034264 8| 6.901724921853395|5.4288799365835 81|3.05
| 27|  Male|            4|  4.607207637203389|6.6840053778031905|   68.92565311534008|6.7788425474858265| 4.512424507733088| 4.285645292447677|4.571380140946013| 2.00548022431093 8|2.3498565973344467| 6.304842194849224| 6.9
| 24|  Male|            4| 2.1865693151850025|  4.01094505664799|   69.14191535245592|   1.854595231091 59|1.1025582818196455|               0.0| 5.98932424248266| 4.02650361258214 1| 4.512920775428038| 4.903146306519509| 5.1
+---+------+-------------+------------------+------------------+-------------------+------------------+------------------+------------------+------------------+------------------+------------------+------
only showing top 5 rows
```

[10]
0s

```
filtered_df = df.filter(col("age") > 23)
filtered_df.show()
```

```
+---+------+-------------+------------------+------------------+-------------------+------------------+------------------+------------------+------------------+------------------+------------------+
|age|gender|academic_year|study_hours_per_day|     exam_pressure|academic_performance|      stress_level|     anxiety_score|  depression_score|      sleep_hours| physical_activity|    social_support|
+---+------+-------------+------------------+------------------+-------------------+------------------+------------------+------------------+------------------+------------------+------------------+
| 29|  Male|            2| 2.580491083921536 3| 6.015297104841322|   58.37236267885964|  3.476176844283 32|2.4252007031850 63|0.8519955424339019|8.946669952792416| 3.29672008034264 8| 6.901724921853395| 5
| 27|  Male|            4|  4.607207637203389|6.6840053778031905|   68.92565311534008|6.7788425474858265| 4.512424507733088|  4.285645292447677| 4.571380140946013| 2.06548022431093 8|2.3498565973344467| 6
| 24|  Male|            4| 2.1865693151850025|  4.01094505664799|   69.14191535245592|  1.854595231091 59|1.1025582818196455|               0.0|  5.98932424248266| 4.02650361258214 1| 4.512920775428038| 4
| 29|Female|            3|  7.697209612629649| 8.460408094949697|   68.33224560138768| 5.464786480217611|1.7203140683193259| 2.3133384217873014| 6.574378981421465|               0.0|3.9281585370590397| 3
| 26|  Male|            4|  6.535580356623476| 6.910608848083312|   77.23699622244354|5.8026085154704035| 4.451598165999091| 2.0345201293455397| 5.157028536295034|2.1388394224398084|3.3408097574019884| 1
| 27|  Male|            2|  6.076286464950613| 6.270351505999681|   69.41441029780003|6.3063586913234735| 5.945700624993356| 1.5120195979375013| 5.691926991114601| 4.65683418983120 1| 7.318289131595173| 8
| 27|Female|            4|   9.50365104744838|  7.58784847529232|   77.13746139416018| 5.235897305800469| 4.324994490754684|  1.5515004467823 69| 8.691459274016909|1.7675121292872782| 4.710295644183864| 2
| 24|  Male|            3| 2.760653007519947 4| 5.969034942406709|   66.17629534354973| 5.408398904349904| 3.882085173245819| 2.604256056244451 7| 4.894768619862015| 4.513007796678313| 5.523283762660319| 6
| 24|  Male|            2|   8.757747727532772| 8.422123741808026|   70.24206831694165| 5.977249844406301|   5.68909562378864| 3.341059972859009 4| 6.602968033734558| 2.67087634989356 2| 2.117406671656004| 4
| 24|Female|            3|  7.688784681574919| 5.809958353127415|   78.62293330514679|2.9128892916540865| 1.521571483377626| 0.9413725885374727| 5.806432995269762| 2.81070500544394 1| 4.781303378263608| 8
| 24|Female|            2|  4.294917707691613| 5.443705961787876|   78.12439543423034| 2.200250023588701|2.2686811052685574|0.36861698711548896|7.474630242756588 5| 4.75506280484585 7| 6.625886119265806| 3
| 28|Female|            4|                 0.0|1.9870813531784044|   60.82873377547772|5.0504382957539615| 4.421448033488448|  2.390093106361552|   4.765330284518 3| 1.31201498783731 2|3.5320095317492792| 6
| 28|Female|            3|  4.260471281169951| 5.509978914993171|    73.1134559474446|1.7589490662543985|0.9941212872736664|                0.0| 5.608554159579941| 5.00285364151995 4| 8.370387766657316|3.
| 28|  Male|            2| 2.354527491035969 3| 4.087351290638844|    73.1352135121601| 2.203619201628198|1.3299571120342275| 1.001023030748196 4| 7.983864830737723|1.5282997800594273|3.484188027746857 3|3.
| 26|Female|            4|  7.319281107550483| 8.549578135827842|    79.5670250412939| 6.414418818688329| 4.029155451747546| 1.2680348247321382|  7.446056519520 87|3.3264386404751924| 6.799436019288107|7.
| 29|  Male|            4|  4.899976129915572| 6.679224724479586|   69.58954179047514|6.8091180691093545|3.8723301073300913| 2.5203770699125174|  4.952505110399962| 5.84539631631835 1| 5.430060460569008|1.
| 28|Female|            4|  4.043613022401594| 4.508314264071062|   75.71860265215712| 4.155052837171907|2.2116975996763744| 2.1247799958730837| 8.290575067832647|0.24573430493113 69|3.3345381364312763| 4
| 25|Female|            4|  6.962244416624089| 7.103702501693533|   78.84735983609204|4.7389072559940875|3.6655101861066903| 1.6583255665315655| 6.834037701138998| 5.39534305957247 9|3.5397082003363662|3.
| 27|Female|            1| 3.0720052503854394| 3.872815269998757|   65.54672193273575|1.5274308516384676|1.3007278310451937|                0.0| 8.705338329210 51|4.77202196173743 95|6.8929364545036105|3.
| 27|Female|            2|  4.328828833046272|  4.67699496979306|   76.99490719134145| 3.381295407451349|1.5187930600977848| 1.0444667829315297| 7.456542893146127| 0.89529027929900 1| 6.742170743198985| 6
+---+------+-------------+------------------+------------------+-------------------+------------------+------------------+------------------+------------------+------------------+------------------+
only showing top 20 rows
```

[11]

[11]
5s

```python
grouped_df = df.groupBy("gender").count()
grouped_df.show()
```

```
+------+------+
|gender| count|
+------+------+
|Female|480070|
| Other| 40287|
|  Male|479643|
+------+------+
```

[12]
5s

```python
sorted_df = grouped_df.orderBy("count", ascending=False)
sorted_df.show()
```

```
+------+------+
|gender| count|
+------+------+
|Female|480070|
|  Male|479643|
| Other| 40287|
+------+------+
```

1. How does Spark divide data?
    - Spark divides data into partitions (logical chunks) that are distributed across worker nodes for parallel processing. The number of partitions is determined automatically based on the cluster resources and input file size.

2. What happens if dataset size increases?
    - When dataset size increases, Spark automatically increases the number of partitions to maintain parallel processing efficiency. More resources (worker nodes) are utilized if available in cluster mode, and data can be spilled to disk if memory limits are exceeded.

3. Why is this better than standard Python processing?

- Spark provides distributed processing across multiple nodes, fault tolerance through task re-execution, horizontal scalability to handle terabytes of data, and optimized execution through lazy evaluation and query optimization.

PART IV: CLOUD COMPUTING ANALYSIS

1. Why is Google Colab considered cloud computing?
    - Google Colab is cloud computing because it provides on-demand computing resources (CPUs, GPUs, memory) from Google's cloud infrastructure, accessible via web browser with no local installation required.

2. Which cloud service model does this resemble (IaaS, PaaS, SaaS)?
    - Google Colab resembles Platform as a Service (PaaS) because it provides a complete development environment with pre-configured tools, while abstracting away the underlying infrastructure management.

3. How does cloud infrastructure help distributed systems?
    - Cloud infrastructure enables distributed systems through elastic scaling, resource pooling, on-demand provisioning, geographic distribution, and managed services that handle node failures and resource allocation automatically.